

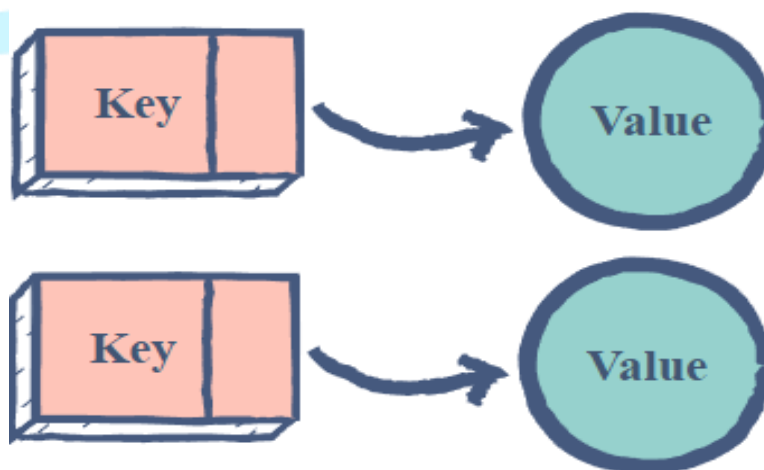


Map Interface in Java:

What is Map?

- A Map in java is a container object that stores elements in the form of key and value pairs.
- Each **key and value pair** are known as an **Entry**.
- The java Map interface is a part of Java Collection Framework, but **it is not a sub-type** of the Collection Interface. So, it behaves in a different way compared to say, Lists or other collection objects.

Key-Value





Additional Information Regarding Map:

- Each entry in the map stores the data in a key and its corresponding value.
- Map interface contains only unique keys and does not allow any duplicates keys but value duplicate.
- Each key maps to only one value. This type of mapping is called **One-To-One Mapping** in java.
- The Map interface in java is a part of the **java.util** package.
- A key is an object that you use to access the value later, it is associated with a single value.
- A map cannot be traversed, therefore you must use the `keySet()` and `entrySet()` method to convert it into a set.

Object Creation of Map Interface:

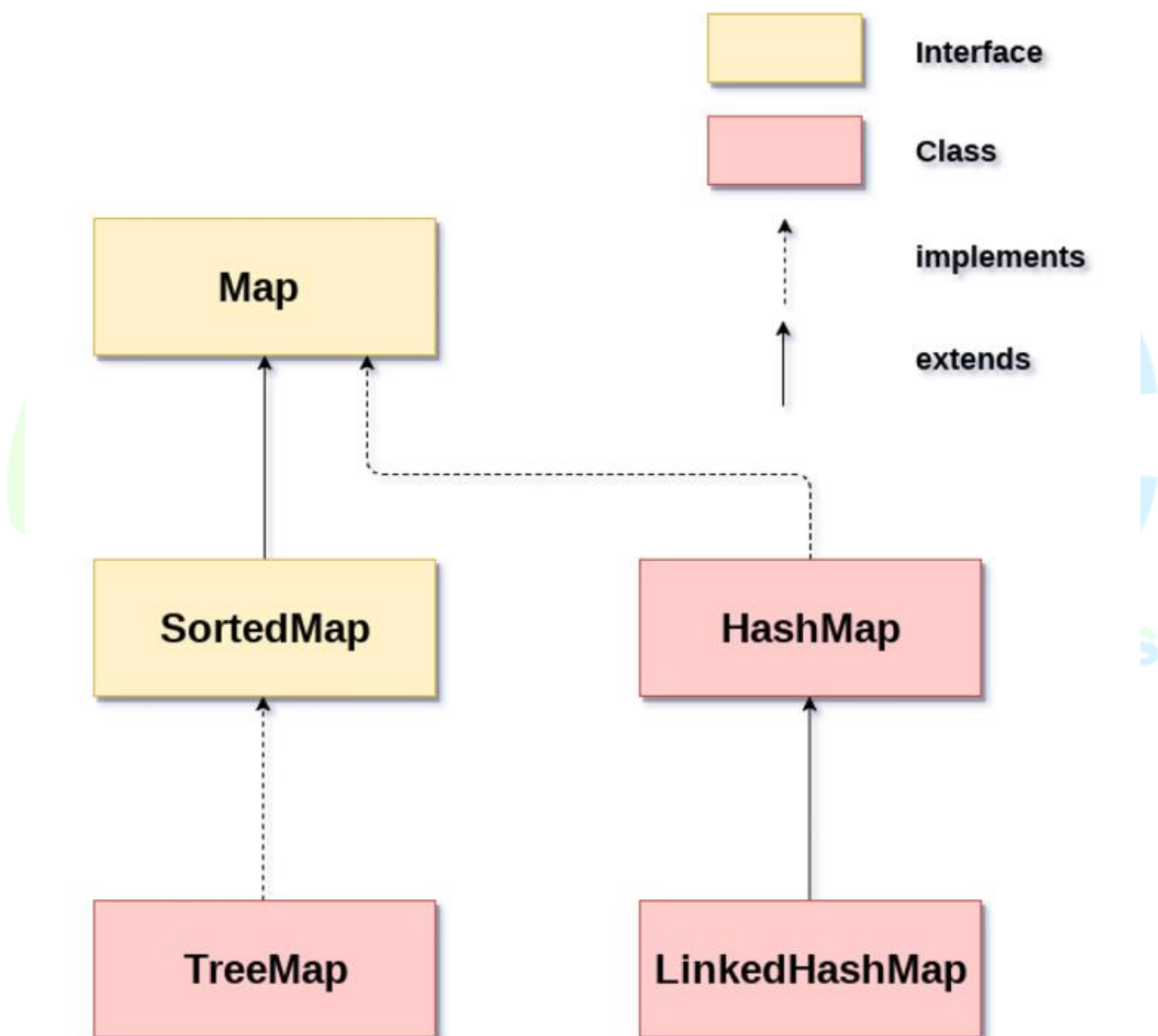
```
Map<K, V> map = new HashMap<K, V>();
```

Where: -

Map	→ Interface.
<K, V>	→ Entry in Map.
map	→ Reference variable / object.
new	→ Keyword.
HashMap	→ Implemented Class.
()	→ Parentheses.
K	→ Key in Map.
V	→ Value in Map.



Hierarchy of Map Interface in Java:





Important Methods of Map Interface:

There are following important methods which is belongs from the Map Interface is given below-

- 1. Put Method (put(k, v)):** - In map, the given value and key are associated together using this method.
- 2. Get Method (get(k)):** - The object containing the value associated with the key is returned by this method.
- 3. Size Method (size()):** - The number of key/value pairs in the map is returned by this method.
- 4. KeySet Method (keySet()):** - It returns Set that contains the key in a map.
- 5. EntrySet Method (entrySet()):** - It returns Set that contains the entries in a map.

Difference between HashMap and Hash Table: -

There are following difference between HashMap and HashTable which is given below –



Hash Map	Hash Table
In hash-map, not a single method is synchronized.	Every method in hash table is synchronized.
In hash-map, multiple threads can run at the same time. Hence it is not thread-safe.	In hash-table, only one thread is allowed to run at a time. Hence it is thread-safe.
In it, thread do not wait, hence performance is high.	On the other hand, threads need to wait which makes the performance low.
Hash-Map does allow the storage of the null values for both key and values.	Hash-Table does not allow null values for both key and value because it will give null pointer exceptions.
Introduced in jdk1.2 version.	Introduced in jdk1.0 version.
It is a non-legacy class.	It is a legacy class.

Difference between HashMap and LinkedHashMap: -

There are following difference between HashMap and LinkedHashMap which is given below –

Hash Map	Linked Hash Map
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Hash Map implements Map interface.	Linked Hash Map extends the HashMap class.
HashMap does not guarantee any order.	LinkedHashMap is based on insertion order.
Hash Map takes less memory.	Linked HashMap consumes relatively more memory.

Difference between HashMap and Hash TreeMap: -

There are following difference between HashMap and TreeMap which is given below –

Hash Map	Tree Map
It does not provide any order for elements.	It provides orders for elements.
It allows one key as null and also allows multiple null values.	It does not allow key as null but it allows multiple null values.
It consumes more memory space.	It consumes less memory space.
It's speed is fast.	I It's speed is low.